

12/3/2025

mercoledì 12 marzo 2025 11:39

Corta e penna

$$\begin{cases} \bar{a} = 1+x \\ \bar{z} = \frac{a-1}{x} = \frac{1+x-1}{x} = \frac{x}{x} = 1 \end{cases}$$

```
>> x = 1.e-15;
```

```
>> a = 1+x;
```

```
>> z = (a-1)/x
```

$$\begin{aligned} a &= 0.147554853 \rightarrow \bar{a} = 0.147555 \\ b &= 0.147551742 \rightarrow \bar{b} = 0.147552 \end{aligned}$$

$$t = 6$$

$$a - b = 0.000003111 = 0.3111 \cdot 10^{-5}$$

$$\bar{a} \ominus \bar{b} = 0.000003 = 0.30000 \cdot 10^{-5}$$

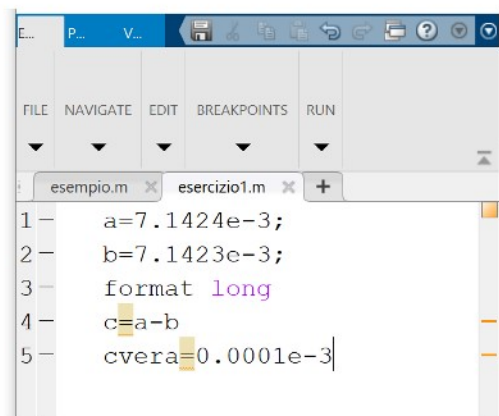
```
>> esercizio1
```

```
c =
```

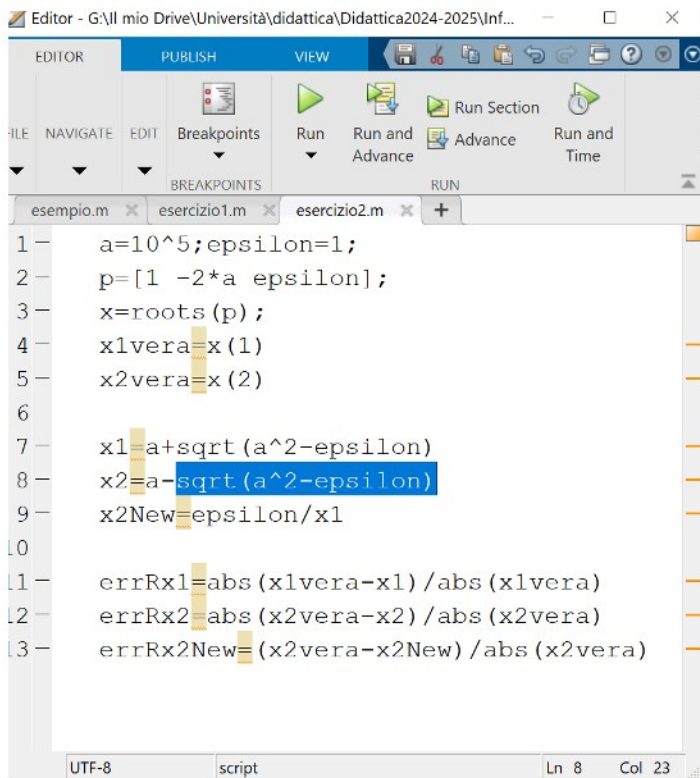
```
1.0000000000002735e-07
```

```
cvera =
```

```
1.0000000000000000e-07
```



```
1- a=7.1424e-3;  
2- b=7.1423e-3;  
3- format long  
4- c=a-b  
5- cvera=0.0001e-3
```



```
1- a=10^5;epsilon=1;
2- p=[1 -2*a epsilon];
3- x=roots(p);
4- x1vera=x(1)
5- x2vera=x(2)
6-
7- x1=a+sqrt(a^2-epsilon)
8- x2=a-sqrt(a^2-epsilon)
9- x2New=epsilon/x1
10-
11- errRx1=abs(x1vera-x1)/abs(x1vera)
12- errRx2=abs(x2vera-x2)/abs(x2vera)
13- errRx2New=(x2vera-x2New)/abs(x2vera)
```

```
>> esercizio2

x1vera =

    1.999999999950000e+05

x2vera =

    5.000000000125000e-06

x1 =

    1.999999999950000e+05

x2 =

    4.999994416721165e-06

x2New =

    5.000000000125000e-06

errRx1 =

    0

errRx2 =

    1.116680766856470e-06

errRx2New =

    0

>> a

a =

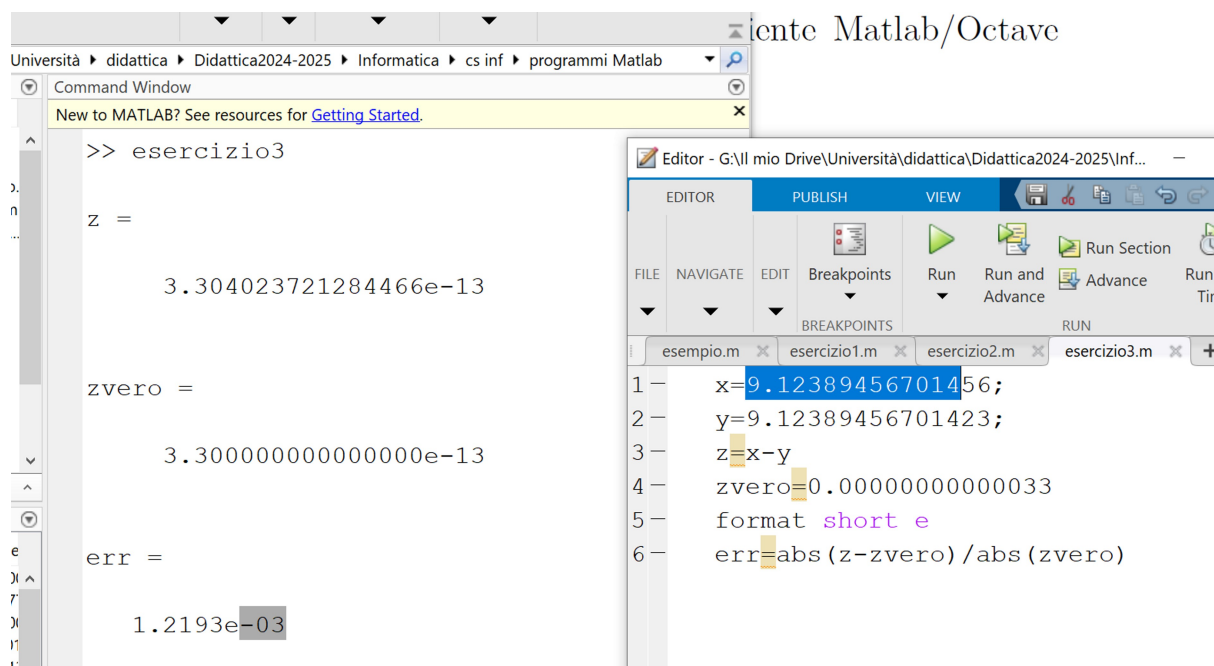
    100000
```

```
>> sqrt(a^2-epsilon)

ans =

9.999999999500001e+04

>>
```



10. Assegnato il valore $x = 10^{15}$, calcolare in ambiente Matlab/Octave l'espressione

$$(1 + 1/x) * x - x.$$

Commenta il risultato ottenuto (confrontando con il risultato ottenuto carta e penna).

$$\left(1 + \frac{1}{n}\right) \cdot x - x = n + \frac{n}{n} - x = \frac{1}{n}$$

Matlab Editor interface showing the Command Window and Variable Editor.

Command Window:

```
>> esercizio4  
  
y =  
  
1.1250000000000000  
  
yvero =  
  
1  
  
err =  
  
1.2500e-01
```

Variable Editor:

Variable	Value
y	1.1250e+01
yvero	1
err	1.2500e-01

Matlab Editor interface showing the Editor window with the script file `esercizio4.m`.

Editor Window:

```
1 x=1e15;  
2 format long  
3 y=(1+1/x)*x-x  
4 yvero=1  
5 format short e  
6 err=abs(y-yvero)/abs(yvero)  
7 format long  
8 %quante cifre hanno in comune i due operandi?  
9 a=(1+1/x)*x  
10 b=x
```